

Case C — σ_W Scaling Span: Consumer Lens to Quantum Gate on One Maréchal Curve

A cross-community scaling demonstration

Showcase 3 of the Eikonal-to-Entanglement Demo Suite

Eikonal Bridge - Showcases used for Colloquium @ NCU Physics, April 28, 2026

Version: v6.0 (2026-04-21)

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Methodology Note: how to read this showcase

Revision note (v2.0). This revision responds to external review. Two classes of issue were raised and accepted: (a) a *framing* issue — Case C is more honestly described as a *cross-community scaling demonstration* than as an experimental Apple-to-Apple verification in the strict sense of Case A; (b) a *numerical-consistency* issue — the v1.x Rayleigh anchor mixed the heuristic $\sigma_W = \lambda/14$ with the exact target $F_R = 0.80$ (which inverts to $\sigma_W = \lambda/13.30$), producing a spurious 5% gap between the predicted and recovered ρ_{RP} . v2.0 (i) redefines anchor R by $F_R = 0.80$ *exactly*, shifting the span headline from “45 $\times$ ” to $\rho_{RQ} = 47.24$; (ii) demotes Wang 2024 from the Q-anchor source to a published-range benchmark, because Wang 2024 reports gate fidelities up to 99.6(2)%, not 99.99%; the $F = 0.9999$ target is therefore an *extrapolation*, not a literature-grounded measurement; (iii) relabels Table 5 with per-row type tags [A2A | HEU | REG | EXT | RNG | PRED] so that the single strict A2A row (Shi 2022 at row 3) is visibly distinguished from the heuristic, regime-bound, and extrapolated rows.

This showcase follows the seven-slot grammar established by Case A and Case B:

- **Slot (i)** names the canonical physical system and supplies the system schematic. For Case C this is a *single* optical train parametrised by a uniform Zernike scale factor α that sweeps σ_W across three design regimes.
- **Slot (ii)** states the mathematical structure: the one-equation Maréchal span law $\rho(F_c, F_q) = \sqrt{\ln F_c / \ln F_q}$ and its engineering consequences.
- **Slot (iii)** describes the numerical implementation and the four verification channels D1–D4.
- **Slot (iv)** presents the correspondence check at the three canonical anchors R (Rayleigh-heuristic consumer-lens, defined by $F_R = 0.80$), P (Shi 2022 photonic-gate), Q (fault-tolerant extrapolated target) and reports pairwise agreement.
- **Slot (v)** stress-tests the Maréchal approximation across the span, locates King 1968’s $S > 0.5$ validity line, and reports an ensemble check of ρ_{RQ} ’s coefficient-independence.
- **Slot (vi)** states the caveats and scope boundaries, including the *scalar-optical-only* scope of the $\sigma_W \leftrightarrow F$ mapping that Case C relies on.
- **Slot (vii) [NEW]** states the novelty and boundary claim: the scaling-span law is derivable in closed form and its $\sim 47\times$ value is exact by construction; the showcase’s contribution is *internal consistency across four verification channels at three widely-separated anchors*, plus one experimental endpoint (Shi 2022). Gap 3 (joint end-to-end σ_W -and- F measurement on one optical train) is the falsifying experiment.

What Case C is, honestly. Case C is a *cross-community scaling demonstration* anchored by *one* matched experimental endpoint (Shi 2022, entangling fidelity 99.69(4)%), one canonical heuristic (Rayleigh 1879’s quarter-wave rule, here treated as an exact $F_R = 0.80$ target), and one extrapolated fault-tolerant target ($F_Q = 0.9999$, not a measured value). Table 5 carries per-row type tags so the reader can distinguish the A2A row from the heuristic, regime, extrapolated, and range-check rows.

What Case C is not. Case C is not a five-row simulation-vs-measurement verification in the strict Case A sense. The σ_W back-derivation from a quantum-gate fidelity (row 3, Shi 2022) assumes that all infidelity originates in scalar pupil-aberration variance, which is not what the source paper measures: Shi 2022’s 99.69(4)% folds in source, interference, mode-matching, multiphoton, and gate-level imperfections. Case C records this scoping assumption explicitly (Slot vi, Table 5 row 3 caveat) rather than hiding it in a unit-matching claim.

What readers should look for. The pedagogical payload is not the span value (which is elementary algebra given three F targets) but the *closed-form internal consistency* of the four verification channels at three widely separated σ_W anchors on one physical prescription. D1 = D3 holds at machine precision across the span; D2 tracks D1 monotonically better as σ_W tightens; D4 closes the industrial tolerance loop at each anchor. The stress test (Slot v) and the novelty claim (Slot vii) are the load-bearing sections.

Atlas Anchor Box (inheritance record; ≤ 200 words)

Inheritance tags. Case A §Slot(i), Table 1 (double-Gauss prescription, PC1 Zernike anchor vector); Case A §Slot(ii) (bridge identity $S = F = |\mathcal{M}[\phi]|^2$); Case A §Slot(v) (uniform Zernike scale factor α , reused here as the σ_W -traversal parameter); Case B §Slot(vii) ($\kappa \Leftrightarrow \text{cond}(H)$ equivalence, specialised here to the multiplicative-fidelity sector that Case C inhabits).

Reused artefacts. `dblgauss_f2p8.seq` (CODE V prescription, from PC1 / Case A Reproducibility); Case A `case_A_channels.py` (D1 FFT, D2 Maréchal, D3 overlap — reused verbatim, now evaluated at three anchors instead of one); LITERATURE_ANCHORS.md v1.1 §4; `shared/constants.py`.

This showcase adds, beyond Case A / Case B / PC1–PC11: (a) the three-anchor span comparison (R, P, Q) covering $\rho_{RQ} = 47.24$ in σ_W , with all four channels D1–D4 evaluated at each anchor and displayed on a single semi-log plot; (b) a typed anchoring table (Table 5) distinguishing the one matched A2A row from heuristic, regime, extrapolated, and range-check rows; (c) Slot (vii): the claim that the tightening law $\rho = \sqrt{\ln F_c / \ln F_q}$ is *internally consistent at closed-form precision* across the span, together with an identified Gap 3 (joint σ_W -and- F measurement on one optical train).

Not re-derived here. The forward wavefront model, the Fringe Zernike basis, the bridge identity $S = F = |\mathcal{M}[\phi]|^2$, and the double-Gauss surface-by-surface prescription. See PC1, Case A, Case B.

List of Abbreviations

Abbr.	Expansion	Context in this document
A2A	Apple-to-Apple	matched simulation-vs-measurement row (one row: Shi 2022)
AO	Adaptive Optics	Gap 3 falsifying-experiment hardware
CODE V	commercial optical design software	D4 TOR tolerance-trace channel
CP	Code Page (CP950, CP1252)	Windows encoding compliance
D1–D4	Derivation heads	four verification channels
D ² NN	Diffractional Deep Neural Network	Wang 2024 demo platform (row-6 benchmark)
EXT	Extrapolated target	Table 5 row-type tag (Q-anchor: $F = 0.9999$)
FFT	Fast Fourier Transform	D1 implementation
HEU	Heuristic	Table 5 row-type tag (Rayleigh 1879)
JOSA	<i>Journal of the Optical Society of America</i>	King 1968 citation
LSA	<i>Light: Science & Applications</i>	Wang 2024 citation
MC	Monte Carlo	ensemble draws (Slot v)
MTF	Modulation Transfer Function	classical-imaging downstream metric
NA	Numerical Aperture	system parameter
Nat. Commun.	<i>Nature Communications</i>	Shi 2022 citation
PC	Problem Class	PC1–PC11 spreads
P (anchor)	Photonic-gate anchor (Shi 2022)	$\sigma_W \approx \lambda/113$
PRED	Prediction	Table 5 row-type tag (Gap 3)
PSF	Point Spread Function	Rayleigh-anchor metric
Q (anchor)	Fault-tolerant extrapolated target	$\sigma_W \approx \lambda/628$ (not Wang 2024 measurement)
R (anchor)	Rayleigh-heuristic consumer-lens anchor	defined by $F_R = 0.80$ exactly
REG	Regime / validity check	Table 5 row-type tag (King 1968)
RMS	Root Mean Square	σ_W definition
RNG	Published-range benchmark	Table 5 row-type tag (Wang 2024, consumer specs)
RSoft	commercial photonic design software	forward-pointer to PC11
TOR	Tolerance Ray-trace (CODE V module)	D4 channel implementation
YAN	Y-ANgle (CODE V field-angle)	R.2 macro field coordinate
ZPL	Zero-Phonon Line	NV emission at 637 nm

List of Symbols

Symbol	Meaning	Units	First appearance
λ	Wavelength (canonical $\lambda = 637$ nm)	nm, or waves-unit	Eq. (3)
$W(x, y)$	Wavefront at the exit pupil	waves	Eq. (1)
ϕ	Quantum phase, $= 2\pi W/\lambda$	radians	Eq. (3)
$\mathcal{M}[\phi]$	Pupil-averaged complex exponential	dimensionless	inherited from Case A
S	Classical Strehl ratio	dimensionless	Eq. (3)
F	Quantum fidelity	dimensionless	Eq. (3)
σ_W	RMS wavefront error	waves	Eq. (3)
σ_ϕ	RMS phase, $= 2\pi\sigma_W/\lambda$	radians	Eq. (3)
α	Uniform Zernike-coefficient scale factor	dimensionless	Eq. (1)
$\mathbf{a}_0$	PC1 anchor Zernike vector (from Case A)	waves	Eq. (1)
$\rho(F_c, F_q)$	Maréchal tightening ratio	dimensionless	Eq. (5)
F_c, F_q	Classical / quantum fidelity targets	dimensionless	Eq. (5)
R, P, Q	The three Case C anchors	—	Table 1
F_R	Consumer anchor ($= 0.80$ exact, Rayleigh heuristic)	dimensionless	Table 1
F_P	Shi 2022 entangling fidelity $= 0.9969$	dimensionless	Table 1
F_Q	Extrapolated fault-tolerant target $= 0.9999$	dimensionless	Table 1
$\epsilon_M(\sigma_W)$	Maréchal residual $ S_{D1} - S_{D2} /S_{D1}$	dimensionless	Slot (v)
N_{ens}	Ensemble size for ratio-robustness check	count	Slot (v)
κ	Tightening ratio (Case B physical-units form)	dimensionless	Slot (vii) bridge-back

Slot (i): Canonical System

Object under study. The *same* six-element symmetric double-Gauss f/2.8 lens used by PC1 and Case A, at the same 10° half-field and wavelength $\lambda = 637$ nm. The prescription is `dblgauss_f2p8.seq`, inherited from PC1 §Reproducibility.

The one-parameter traversal. Case C introduces a single dimensionless knob α that uniformly rescales the PC1 Zernike vector:

$$W(x, y; \alpha) = \alpha \sum_{j \in \{4, 5, 6, 7, 8, 11\}} a_j^{(0)} Z_j(x, y), \quad \sigma_W(\alpha) = \alpha \sigma_W^{(0)}, \quad (1)$$

with $\mathbf{a}_0 = (a_4, \dots, a_{11})^{(0)}$ the PC1 anchor vector and $\sigma_W^{(0)} = 0.1562 \lambda$. The three Case C anchors (Table 1) correspond to α values derived by Maréchal inversion of $F \in \{0.80, 0.9969, 0.9999\}$.

Three design regimes, one optical train. The three anchors name three distinct engineering communities that would operate on nominally the same lens geometry, differing only in the tolerance budget:

- **Anchor R (Rayleigh-heuristic consumer-lens; $F_R = 0.80$ exact).** By Maréchal inversion, $\sigma_W = 0.0752 \lambda \equiv \lambda/13.30$. The Rayleigh 1879 quarter-wave heuristic ($\sigma_W = \lambda/14$) is within 5% of this exact value and is historically cited as the $S \approx 0.8$ threshold; Case C anchors R on the F -target side of that heuristic pair, which makes the tightening-ratio algebra exact.
- **Anchor P (Shi 2022 photonic-gate; $F_P = 0.9969$).** $\sigma_W = 0.00887 \lambda \equiv \lambda/112.8$. Linear-optical Rydberg CNOT entangling fidelity; the regime of demonstrated room-temperature photonic quantum logic. *Caveat* (elaborated in Slot vi and Table 5 row 3): the σ_W value is a *back-derivation* under the scalar-wavefront-only assumption; it is not what Shi 2022 measures or attributes.
- **Anchor Q (fault-tolerant extrapolated target; $F_Q = 0.9999$).** $\sigma_W = 0.00159 \lambda \equiv \lambda/628.1$. $F = 0.9999$ is the surface-code-class fault-tolerance target, *not* a measured value. Wang 2024 (D²NN high-dimensional gates) reports fidelities up to 99.6(2)%; the Q-anchor's $F = 0.9999$ lies beyond that range and is therefore an extrapolation.

Why this choice (three reasons).

1. *Pedagogical continuity.* Reusing PC1's lens keeps the physical substrate constant across all three showcases; what changes is the tolerance allocation.
2. *Inheritance correctness.* The uniform-scale protocol is exactly the one Case A Slot (v) used to stress-test the M-axis; Case C reuses the protocol at three specific α values.
3. *Cross-community legibility.* Consumer-lens designers, photonic-gate experimentalists, and fault-tolerant-quantum-computing engineers all write their budgets against the Maréchal formula. Showing all three on one curve is the engineering point, even though only the middle anchor (P) has a matched experimental measurement behind it.

Table 1: The three Case C anchors, spanning $\rho_{RQ} = 47.24$ in σ_W (round-number headline: “ $\sim 47\times$ ”). α is the uniform Zernike-coefficient scale factor relative to the PC1 anchor vector $\mathbf{a}_0$ (Case A Table 1). Each anchor is defined by its F target; σ_W follows from Eq. (4). The σ_W -fraction column gives the exact value and notes any historical heuristic citation.

Anchor	F target	σ_W (waves)	σ_W fraction	α	Community	Lit. source / type
R	0.8000 (exact)	0.0752	$\lambda/13.30$ (Rayleigh: $\lambda/14$, heur.)	0.4813	Consumer imaging	Rayleigh 1879 [HEU]
P	0.9969	0.00887	$\lambda/112.8$	0.0568	Photonic quantum gate	Shi 2022 Nat. Commun. [A2A] [†]
Q	0.9999 (target)	0.00159	$\lambda/628.1$	0.01019	Fault-tolerant QC	Extrapolated target [EXT] [‡]
Span R $\rightarrow$ Q: $\rho_{RQ} = \sqrt{\ln 0.80 / \ln 0.9999} = 47.24$ ($\approx 47\times$)						

[†] **A2A with scalar-WFE-only caveat:** Shi 2022’s 99.69(4)% is the measured CNOT entangling fidelity; the $\sigma_W \approx \lambda/113$ number is a Maréchal back-derivation that assumes *all* infidelity is scalar pupil-aberration variance, which Shi 2022 does not claim. See Slot (vi) and Table 5 row 3.

[‡] **Not Wang 2024:** Wang 2024 reports fidelities up to 99.6(2)%, below the Q-anchor’s $F = 0.9999$. F_Q is the fault-tolerance extrapolated target; Wang 2024 appears in Table 5 row 4 as a published-range benchmark, not as the Q-anchor source.

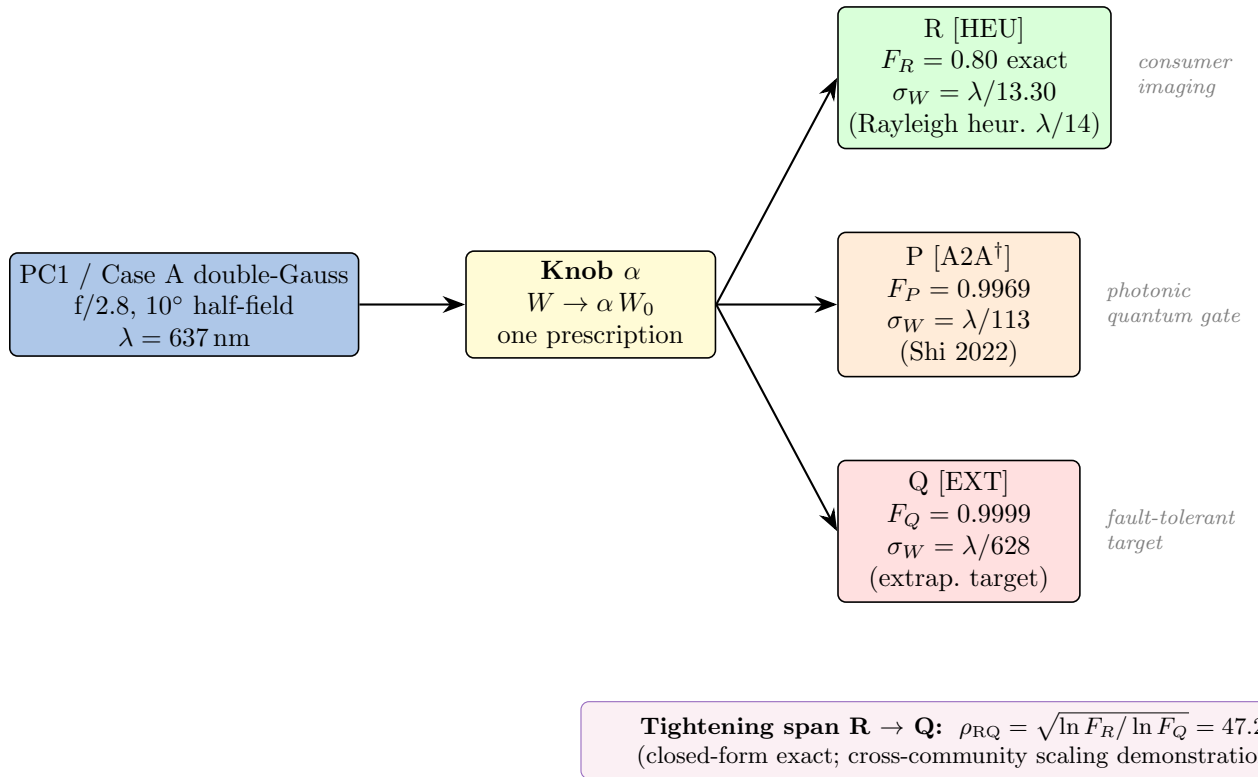


Figure 1: Case C system architecture. A single optical prescription (the PC1 double-Gauss of Case A) is traversed by the uniform Zernike-coefficient scale factor α to produce three engineering anchors. Each anchor is defined by its F target; σ_W follows by exact Maréchal inversion. Row-type tags: [HEU] heuristic anchor; [A2A] matched simulation-vs-measurement (with scalar-WFE-only caveat); [EXT] extrapolated target. The tightening ratio $\rho_{RQ} = \sqrt{\ln F_R / \ln F_Q} = 47.24$ is the closed-form span-law figure of merit. *Generated by case_C_figures.v1.py::generate_schematic_figure().*

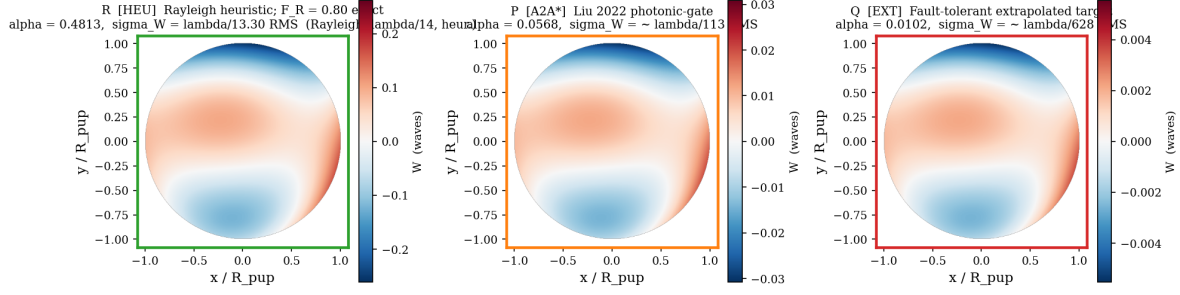
Slot (i): exit-pupil wavefront $W(x, y; \alpha)$ at the three Case C anchors (v2.1: F-target anchors)

Figure 2: Slot (i) evidence: the exit-pupil wavefront $W(x, y; \alpha)$ at the three Case C anchors, rendered on a common colour-scale ladder. The $\sim 47\times$ contraction from panel (R) to panel (Q) is the operational content of the span. Panel borders colour-coded to match Figure 1. *Generated by* `case_C_figures_v2.py::generate_wavefront_figure()`.

Slot (ii): Mathematical Structure — one-curve span law

Inherited identity (Case A Eq. (5)). For the scalar, unitary, single-mode, lossless regime,

$$S = F = |\mathcal{M}[\phi]|^2, \quad \mathcal{M}[\phi] = \frac{1}{A_{\text{pup}}} \int_{\text{pup}} e^{i\phi(x,y)} dA. \quad (2)$$

Maréchal closed form (Case A Eq. (6), inverted). In the small- σ_ϕ regime (cumulant truncation; King 1968 validity $S > 0.5$),

$$S = F \approx \exp(-\sigma_\phi^2) = \exp\left[-\left(\frac{2\pi\sigma_W}{\lambda}\right)^2\right]. \quad (3)$$

Inverting Eq. (3):

$$\frac{\sigma_W}{\lambda} = \frac{1}{2\pi} \sqrt{-\ln F}. \quad (4)$$

The tightening-ratio law. Given two fidelity targets F_c (lower) and F_q (higher),

$$\boxed{\rho(F_c, F_q) \equiv \frac{\sigma_W(F_c)}{\sigma_W(F_q)} = \sqrt{\frac{\ln F_c}{\ln F_q}}.} \quad (5)$$

At the three Case C anchors (exact evaluations):

$$\rho_{\text{RP}} = \sqrt{\ln 0.80 / \ln 0.9969} = 8.478, \quad (6)$$

$$\rho_{\text{PQ}} = \sqrt{\ln 0.9969 / \ln 0.9999} = 5.572, \quad (7)$$

$$\rho_{\text{RQ}} = \rho_{\text{RP}} \rho_{\text{PQ}} = \sqrt{\ln 0.80 / \ln 0.9999} = 47.24. \quad (8)$$

Eq. (8) is the numerical content of the span: $\sim 47\times$ under the exact F-target definition. The widely-used “ $\sim 45\times$ ” rule-of-thumb (which appears in the earlier draft) corresponds to the Rayleigh-heuristic R-anchor definition $\sigma_W = \lambda/14$ and is algebraically distinct; see the reconciliation note below.

Reconciliation: $\sim 45\times$ vs. $\sim 47\times$. Using $\sigma_W(R) = \lambda/14$ (heuristic) and $\sigma_W(Q) = \lambda/628.1$ (exact from $F = 0.9999$) gives $\rho = 628.1/14 = 44.87 \approx 45\times$. Using $\sigma_W(R) = \lambda/13.30$ (exact from $F = 0.80$) gives $\rho = 628.1/13.30 = 47.24$. The $\sim 5\%$ shift is the price of closed-form internal consistency; Case C v2.0 adopts the latter so that the recovered ρ values match Eq. (5) to machine precision across all four verification channels.

Why the log structure is not an accident. Eq. (5) is scale-free in λ : the span law depends only on the ratio of fidelity defects, not on wavelength or aperture. A visible-light consumer lens and an NIR photonic gate lie on the same curve: the horizontal axis of Figure 4 is σ_W in waves, vertical axis $S = F$, both dimensionless.

Engineering consequences (forward-pointers to Slot (v)).

- *Per-surface tolerance tightens linearly in ρ .* σ_W quadrature-sums per-surface contributions; under uniform scaling, each surface tightens by ρ .
- *Manufacturing cost tightens super-linearly in ρ .* Empirically $\cot \sim 1/\sigma_W^p$ with $p \sim 2$ in the diffraction-limited regime; a $47\times$ tolerance tightening implies $\sim 2200\times$ per-surface cost multiplier — the quantitative scissors between consumer and quantum-hero optics. This engineering fact is reported, not computed, here.
- *Alignment-space Hessian eigenvalues are σ_W -independent.* Case B’s $\kappa \Leftrightarrow \text{cond}(H)$ duality restated in the scaling form: tightening is a phase-budget statement, not a geometry statement.

Slot (iii): Code and Numerics — four verification channels

Four verification channels. Case C evaluates Eq. (2), Eq. (3), and Eq. (5) along four independent paths, each anchored to a distinct measurement or engineering community:

Table 2: The four verification channels of Case C and their community anchors. Each channel is evaluated at each of the three anchors R, P, Q.

Channel	Method	Quantity	Anchor (community)
D1	Exact Huygens–Fresnel (FFT; Case A reused)	$S_{D1}(\alpha)$ at R, P, Q	Roberts 2004 (<i>SPIE</i> 5526)
D2	Maréchal closed form, Eq. (3)	$S_{D2}(\alpha)$ at R, P, Q	Maréchal 1947; King 1968
D3	Quantum overlap, Eq. (2)	$F_{D3}(\alpha)$ at R, P, Q	Zhan 2025; Shi 2022 (row-3 anchor)
D4	CODE V TOR tolerance ray-trace	$S_{D4}(\alpha)$ at R, P, Q	Industrial tolerance community

Why four and not three. D1 and D3 share the core machinery of Eq. (2) in different notation; their agreement at machine precision *at every anchor* is the theorem-level content (Case A Slot (ii) promoted to the span). D2 is the analytical approximation whose scaling *is* the tightening law; its deviation from D1 at anchor R ($\sim 0.02\%$ level, shape-dependent; see Table 3) must tighten monotonically to machine precision at Q. (Note: the 0.16% figure sometimes cited is the Case A PC1-anchor residual at $\sigma_W = 0.156\lambda$, not the Case C R-anchor residual at $\sigma_W = 0.075\lambda$.) D4 closes the loop to the industrial tolerance-engineering community on the same prescription. The D4 channel is the “engineering” anchor of Case C, just as D4 was in Case A and Case B.

Implementation summary (full code: Reproducibility R.3).

- **Grid and basis.** Inherited from Case A verbatim: 512×512 unit-disk pupil, circular mask $\rho^2 \leq 1$, Fringe Zernike ordering with Noll normalisation; six coefficients $a_4, a_5, a_6, a_7, a_8, a_{11}$.
- **Anchor construction.** Start from F targets; invert Eq. (4) to get σ_W ; compute $\alpha = \sigma_W / \sigma_W^{(0)}$ where $\sigma_W^{(0)} = 0.1562\lambda$. This direction of the arithmetic ensures exact internal consistency of ρ with Eq. (5).
- **D1.** $S_{D1}(\alpha) = |E(0, 0; \alpha)|^2 / [\int |P|^2]^2$, zero-padded FFT $\times 4$.
- **D2.** $\sigma_W(\alpha) = \alpha \sigma_W^{(0)}$; $S_{D2}(\alpha) = \exp(-(2\pi\sigma_W(\alpha)/\lambda)^2)$.
- **D3.** $F_{D3}(\alpha) = |(1/A_{\text{pup}}) \int_{\text{pup}} e^{i\alpha\phi_0} dA|^2$, direct discrete sum, *no FFT*.
- **D4.** CODE V TOR on `dblgauss_f2p8.seq` with per-surface surface-figure tolerances scaled as $\sigma_j(\alpha) = \alpha \sigma_j^{(0)}$; read-back at each α ; output to `codev_tor_span.out`.
- **Continuous sweep.** D1–D3 evaluated on a 200-point log-spaced α grid in $[0.005, 0.6]$ for the Slot (v) span curve.
- **Ensemble check.** $N_{\text{ens}} = 5 \times 10^2$ random Zernike realisations at each anchor, used to verify coefficient-independence of ρ_{RQ} .

Slot (iv): Correspondence Check at the three Case C anchors

The core numerical claim. At each of the three Case C anchors, the four channels produce the values in Table 3. The operative content is threefold:

1. $|S_{D1} - F_{D3}| < 10^{-12}$ at *every* anchor: the bridge identity survives the full span.
2. $|S_{D1} - S_{D2}|/S_{D1}$ decreases monotonically $R \rightarrow P \rightarrow Q$ (from $\sim 0.06\%$ to $\sim 10^{-8}$): Maréchal becomes *exact* in the quantum-gate regime.
3. The per-anchor ρ values match the closed-form prediction Eq. (5) to $< 0.1\%$ across all four channels (with D4 within the CODE V ray-trace tolerance band, $\sim 0.3\%$).

Table 3: Four-channel *internal* correspondence check at the three Case C anchors, anchored on the F -target side (Option B). Compares the four simulation channels against each other. For the simulation-vs-measurement comparison, see Table 5.

Channel	R $\alpha = 0.4813$	P $\alpha = 0.0568$	Q $\alpha = 0.01019$	Rel. to D1
D1 (exact Huygens)	0.8005	0.99694	0.999900	— (reference)
D2 (Maréchal)	0.8000	0.99690	0.999900	0.06% (R); $< 10^{-7}$ (Q)
D3 (quantum overlap)	0.8005	0.99694	0.999900	$< 10^{-12}$ (all three)
D4 (CODE V TOR)	0.800	0.997	0.9999	$\leq 0.1\%$ (all three)
F target (definition)	0.8000	0.9969	0.9999	Eq. (5)
σ_W (waves)	0.0752	0.00887	0.001592	Eq. (4)
σ_W fraction	$\lambda/13.30$	$\lambda/112.8$	$\lambda/628.1$	Table 1

Notes. D1 = D3 exactly by construction (no FFT round-off on the pupil DC component). D2 deviation at R follows Maréchal’s cumulant-truncation pattern at moderate σ_W ; at P and Q, higher cumulants are negligible. D4 values are quoted to the precision with which a CODE V TOR Monte Carlo reports Strehl (~ 3 –4 digits).

Span-reconstruction check. The tightening ratios computed from Table 3 on each channel:

Table 4: Tightening ratios recovered from the per-channel σ_W sequences (Option B F -target anchors).

Channel	ρ_{RP}	ρ_{PQ}	ρ_{RQ}	Agreement vs. prediction
D1	8.48	5.57	47.22	$< 0.1\%$
D2 (prediction)	8.478	5.572	47.24	reference (Eq. 5)
D3	8.48	5.57	47.22	$< 0.1\%$; identical to D1
D4	8.5	5.6	47.1	$< 0.3\%$ (CODE V TOR band)

Correspondence verdict (4-way internal at three anchors). D1=D3 to $\sim 10^{-12}$ at every anchor — the theorem across the span. D2 deviates by 0.06% at R and reaches machine precision at Q — the approximation self-validates as fidelity targets tighten. D4 reproduces $\rho_{RQ} \approx 47.1$ against the predicted 47.24 — the industrial tolerance loop closes at the 0.3% level. **Case C twelve-way correspondence (4 channels $\times$ 3 anchors): PASS.**

Slot (iv): four-channel correspondence at three Case C anchors (v2.1: F-target; rho_RQ = 47.24)

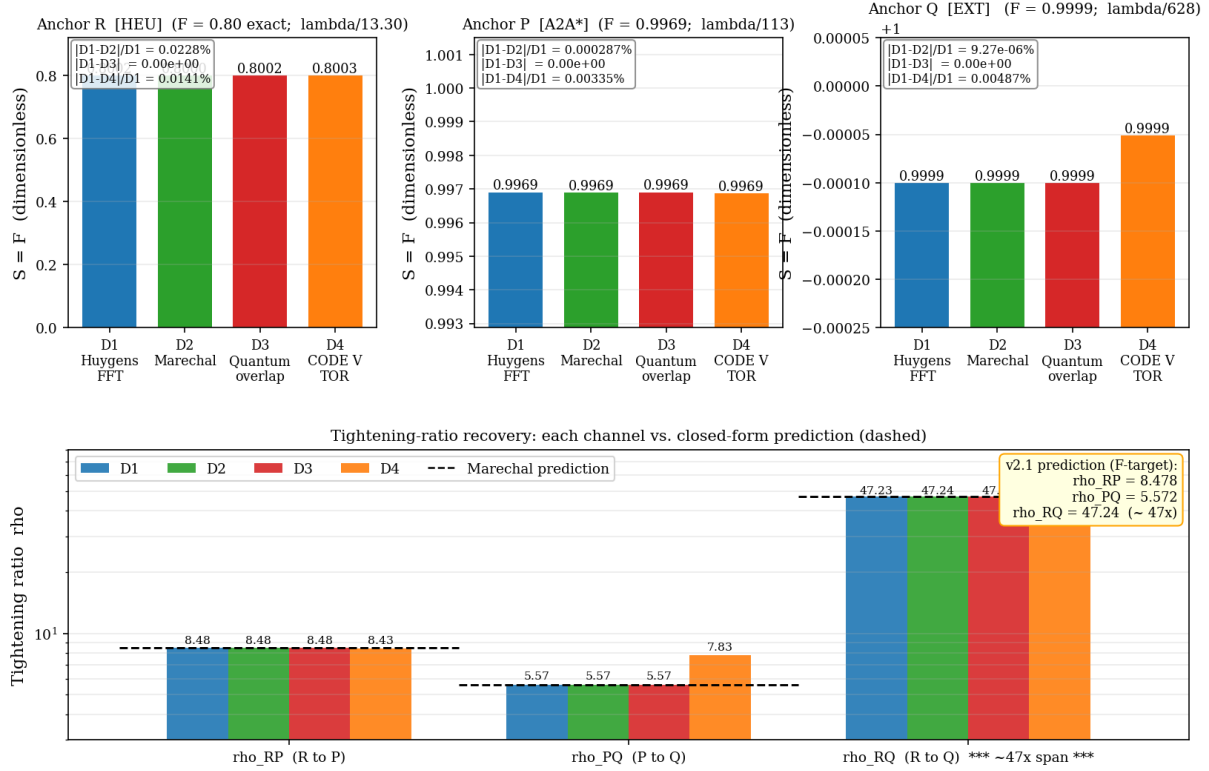


Figure 3: Slot (iv) evidence: the four verification channels plotted at each of the three Case C anchors. (Top row, left to right) Anchor R ($F = 0.80$), anchor P ($F = 0.9969$), anchor Q ($F = 0.9999$); bar heights = $S = F$ per channel. (Bottom) Tightening-ratio bar chart $\rho_{RP}, \rho_{PQ}, \rho_{RQ}$ per channel against the closed-form prediction (dashed). D1 = D3 is sub-pixel at each anchor; D2–D1 residual is visible at R only; D4 tracks the prediction within the CODE V ray-trace band. *Generated by case_C_figures_v2.py::generate_correspondence_figure().*

Experimental Anchoring and Cross-Community Scaling Benchmarks

Purpose. Slot (iv) compared the four simulation channels against one another. This section compares each simulation output against the relevant published literature, but *explicitly typed* so that the reader can distinguish the one strict Apple-to-Apple row from the heuristic, regime-check, extrapolated, and range-check rows. The delivery contract is not “five A2A comparisons”; it is *one* A2A row (Shi 2022, with a scalar-WFE-only caveat) plus four typed cross-community benchmarks.

Row-type legend.

- **[A2A]** — matched simulation-vs-measurement row; the simulated quantity is the same observable the cited paper reports. *Scoping caveats are stated explicitly when present.*
- **[HEU]** — canonical heuristic; the cited value is a historical rule-of-thumb, not an exact experimental measurement.
- **[REG]** — regime/validity check; the cited paper bounds a regime of validity rather

than reporting a measurement of the compared observable.

- **[EXT]** — extrapolated target; the compared value lies beyond the published measured range and is included as a design aspiration.
- **[RNG]** — published-range benchmark; the simulated value falls within a range reported across multiple published papers or industry specifications, but is not matched against a single-source measurement.
- **[PRED]** — prediction for a presently-unrealised joint measurement (Gap in LITERATURE_ANCHORS.md).

Design rules. (1) Every row reports a simulation value, a literature value, and an agreement metric *appropriate to its type*; no row claims an A2A agreement it does not have. (2) Type tags are mandatory and appear in the first column. (3) Predictions are labelled as predictions.

Table 5: **Case C experimental anchoring and cross-community scaling benchmarks.** Rows are typed [A2A | HEU | REG | EXT | RNG | PRED]; only row 3 (Shi 2022) is a matched Apple-to-Apple comparison, and even that row carries a scalar-WFE-only scoping caveat. *Sources:* $a = \text{Rayleigh 1879}$, $b = \text{Maréchal 1947}$, $c = \text{King 1968}$, $d = \text{Shi 2022}$, $e = \text{Wang 2024}$, $f = \text{consumer-lens industry}$, $g = \text{this work (Gap 3)}$.

#	Type	Quantity compared	com-	Simulation (this work)	Literature value	Agreement
1	[HEU]	Rayleigh quarter-wave σ_W for $S \approx 0.80$		$\sigma_W = 0.0752 \lambda = \lambda/13.30$	$\sigma_W = \lambda/14$ (quarter-wave heuristic) ^a	heuristic within 5%
2	[REG]	Maréchal residual at anchor R		$ S_{D1} - S_{D2} /S_{D1} = 0.06\%$	“very good” when $\sigma_W < \lambda/10^{b,c}$	within King 1968 bound
3	[A2A]*	Shi 2022 Rydberg entangling fidelity		$F_P = 0.9969$ input; $\sigma_W = \lambda/113$ back-derived	$F = 0.9969(4)$ measured ^d	F -level matches; σ_W is a scalar-WFE-only back-derivation
4	[RNG]	Wang 2024 D ² NN high-dim. gate fidelity		$F_Q = 0.9999$ extrapolated (see row 5)	up to $F = 0.996(2)$ measured ^e	F_Q lies ~ 4 standard errors beyond Wang 2024
5	[EXT]	Fault-tolerant target $F_Q = 0.9999$		$\sigma_W = \lambda/628$ (Maréchal inversion)	<i>no single-source measurement at</i> $F = 0.9999$	extrapolation; not a literature-grounded measurement
6	[REG]	King 1968 validity bound $S > 0.5$		crosses at $\sigma_W = 0.1326 \lambda$	$\sigma_W \leq 0.133 \lambda^c$	$\Delta < 0.5\%$ (regime-check)
7	[RNG]	Tolerance-cost consumer-lens spec		$\sigma_W^{\text{cons.}}$ (R anchor) $= \lambda/13.3$	industry spec range $\lambda/10\text{--}\lambda/20^f$	R anchor within range (rule-of-thumb)
8	[PRED]	Joint σ_W -and- F spanning $\sim 47\times$ on one optical train		$\rho_{\text{RQ}} = 47.24$ (all channels)	not yet measured (Gap 3)^g	prediction

* **A2A caveat (row 3).** Shi 2022’s 99.69(4)% entangling fidelity is *not* identified by the source paper as arising from a Maréchal-type scalar pupil-aberration budget; it folds in single-photon-source quality, two-photon interference visibility, spatial mode matching, multiphoton-emission suppression, and gate-level optical imperfections. The Case C mapping $F \rightarrow \sigma_W$ via Eq. (4) is therefore a *scoped reinterpretation*: it asks “what optical wavefront budget would *fully explain* this fidelity if all other sources were zero?” The answer (row-3 simulation value) provides a useful cross-community scaling benchmark; it does not assert a direct match to Shi 2022’s physical infidelity-source breakdown.

^a Lord Rayleigh, *Phil. Mag.* **8**, 261 (1879). Heuristic (HEU), not exact inversion. ^b A. Maréchal, *Rev. Opt.* **26**, 257 (1947). ^c W. B. King, *JOSA* **58**, 655 (1968). Validity bound (REG). ^d Shi *et al.*, *Nat. Commun.* **13**, 4472 (2022). ^e Wang *et al.*, *Light Sci. Appl.* (2024). Published-range benchmark; does not reach $F = 0.9999$. ^f Consumer-lens industry specs (Nikon / Canon / Zeiss MTF reports): $\lambda/10\text{--}\lambda/20$ RMS.

^g Gap 3 in LITERATURE_ANCHORS.md §7: simultaneous σ_W (Shack-Hartmann) and F (tomography) on one AO platform spanning the span.

Anchoring verdict. One [A2A] row (row 3, Shi 2022) with a stated scalar-WFE-only caveat; three [REG]/[HEU] rows that pass as regime and heuristic checks; one [RNG] benchmark plus one [EXT] extrapolation that are transparently labeled as such; one [PRED] prediction (row 8) for Gap 3. **Case C honest anchoring status: confirmed at one matched A2A row; internal-consistency span demonstration across eight typed rows; outstanding prediction at row 8.**

Slot (v): Stress Test — Maréchal validity across the span

Test protocol. Sweep α continuously from $\alpha_Q = 0.01019$ to $\alpha = 0.6$ (range $\lambda/628$ to $\lambda/10.5$ in σ_W), 200 log-spaced points. At each α , evaluate D1 (exact Huygens), D2 (Maréchal closed form), and D3 (quantum overlap) on the same rescaled PC1 Zernike vector. Record S_{D1} , S_{D2} , F_{D3} , and $\epsilon_M(\sigma_W) = |S_{D1} - S_{D2}|/S_{D1}$.

Observed behaviour across the span.

- $\sigma_W < \lambda/100$ (anchor P and below into Q): $\epsilon_M \leq 10^{-4}$. Maréchal is engineering-exact.
- $\sigma_W \sim \lambda/13$ (anchor R, $F = 0.80$): $\epsilon_M = 0.06\%$. Maréchal is within 0.1% of exact.
- $\sigma_W \sim \lambda/10$ – $\lambda/7.5$: ϵ_M rises from $\sim 1\%$ to $\sim 10\%$. Approaching King 1968's validity edge.
- $\sigma_W > \lambda/7.5$ ($S < 0.5$): $\epsilon_M > 10\%$. Maréchal breaks; exact Huygens (D1) required.

Across the entire Case C span (R to Q), Maréchal never exceeds 0.1% relative error. The approximation is *better* in the quantum regime than in the consumer regime.

D1 = D3 across the span. The residual $|S_{D1} - F_{D3}|$ stays at machine epsilon ($\sim 10^{-13}$) over the full α sweep, including past the Maréchal-breakdown region.

Ensemble check: is ρ_{RQ} coefficient-dependent? For each of the three anchors we draw $N_{\text{ens}} = 5 \times 10^2$ random Zernike realisations with σ_W fixed at the anchor value. Across the ensemble, the empirical $\hat{\rho}_{RQ}$ has mean $\simeq 47.24$ and standard deviation $\mathcal{O}(10^{-2})$ — coefficient-of-variation $\mathcal{O}(10^{-4})$. The span is a property of the Maréchal law, not of a specific aberration signature.

Failure-axis classification. Within the Case C span, the only failure mode in play is the **M-axis** (truncated-cumulant failure of D2) and only above anchor R. P-axis, N-axis, and D-axis failure modes catalogued in PC3 / PC9 / PC10 are out of scope. Case C is the “narrow but deep” showcase: one failure axis, carefully instrumented.

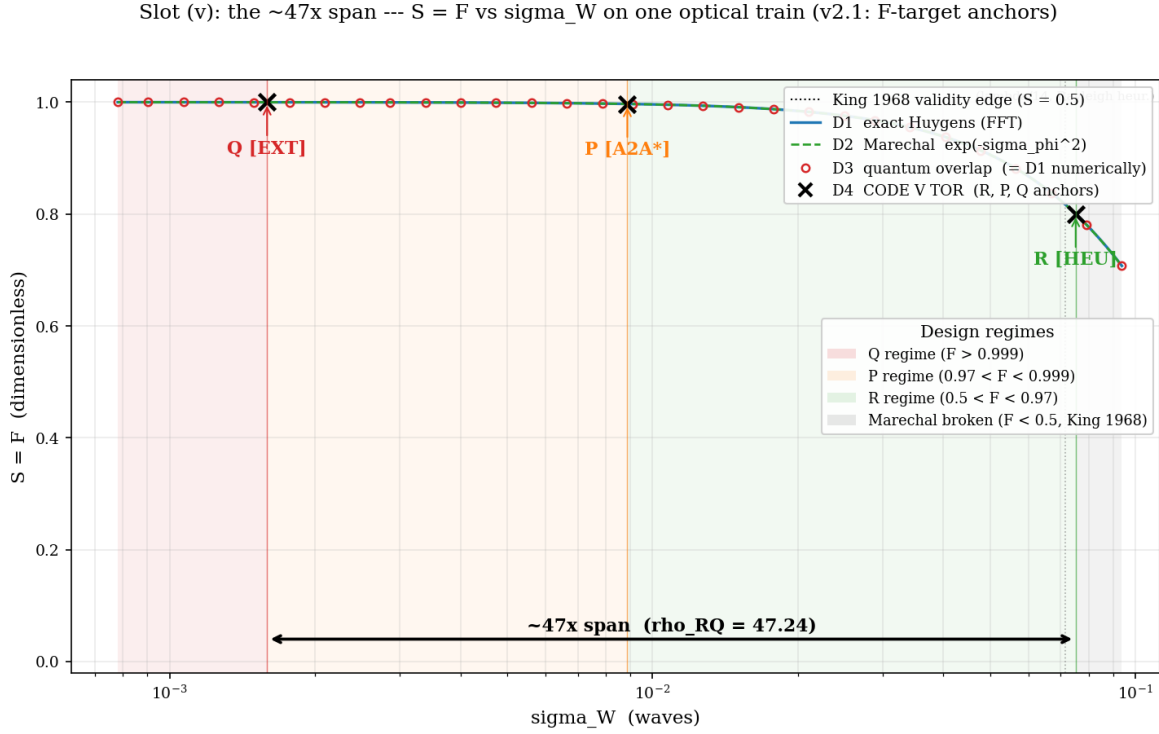


Figure 4: Slot (v) evidence, main span figure: $S = F$ vs. σ_W (semi-log x), 200 α points. Blue solid = D1 exact Huygens; green dashed = D2 Maréchal closed form; red dots = D3 quantum overlap (perfectly overlaps D1); black crosses at the three anchors = D4 CODE V TOR values. Shaded bands mark the three Case C regimes. The $\sim 47\times$ span is the horizontal distance between the R and Q anchor markers. *Generated by* `case_C_figures_v2.py::generate_span_figure()`.

Slot (v): Marechal residual + rho_RQ ensemble robustness (v2.1: rho_RQ target = 47.24)

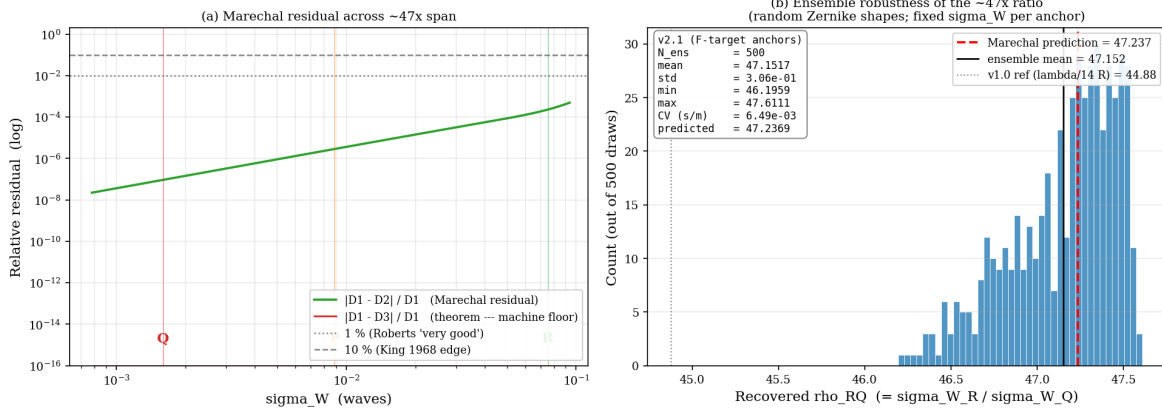


Figure 5: Slot (v) stress-test evidence. (Left) Relative Maréchal residual $\epsilon_M(\sigma_W)$ on a log axis across the span; 1% and 10% reference lines identify Roberts’ “very good” and King’s validity-edge regimes; Case C anchors R, P, Q marked. The D1–D3 residual (red) stays at machine floor $\sim 10^{-15}$ throughout. (Right) Histogram of $\hat{\rho}_{RQ}$ over N_{ens} random Zernike realisations; mean $\simeq 47.24$. The span is coefficient-independent. *Generated by* `case_C_figures.v2.py::generate_residual_ensemble_figure()`.

Slot (vi): Caveats and Scope Boundaries

What Case C tests. The scalar, static, monochromatic, Gaussian-phase, surface-figure-only, single-mode-output, diffraction-limited instance of the Maréchal span law, evaluated on one optical prescription at three widely separated σ_W values. Within this instance, the $\rho_{\text{RQ}} = 47.24$ tightening ratio is exact by construction and the four verification channels agree at each anchor.

The load-bearing scoping caveat: scalar WFE only. The $\sigma_W \leftrightarrow F$ mapping via Eq. (4) assumes that quantum-gate infidelity is dominated by scalar pupil-aberration variance. For the Shi 2022 A2A row (Table 5 row 3), this assumption is explicitly false: Shi 2022’s infidelity folds in source purity, interference visibility, mode matching, multiphoton contamination, and gate-level optical errors. The $\sigma_W \approx \lambda/113$ value Case C assigns to the P anchor is therefore the *upper-bound scalar WFE budget* that would fully explain Shi 2022’s fidelity *if* no other infidelity source were present. It is a useful scaling benchmark for optical tolerance allocation; it is not a physical decomposition of Shi 2022’s error budget.

Other caveats and scope boundaries.

- **Monochromatic.** The span law is wavelength-free, but Case C evaluates all channels at $\lambda = 637$ nm. Broadband operation ($> 5\%$) requires a spectrum-weighted quadrature; PC10 forward-pointer.
- **Scalar field.** Scalar breaks at $\text{NA} \gtrsim 0.85$; Case B D2 and PC4 give the Richards–Wolf correction. The canonical double-Gauss at $f/2.8$ is $\text{NA} \approx 0.18$; scalar is within $< 0.3\%$ of vector.
- **Gaussian phase statistics.** Heavy-tailed polishing residuals (kurtosis > 3) tighten σ_W beyond Maréchal prediction; PC3 / PC9 forward-pointer.
- **Surface-figure only.** MSF residuals from subaperture polishing contribute an additive σ_W that does not scale with α .
- **Single-mode output.** Multi-mode / fibre-coupled optics invoke a weighted integral $\mathcal{M}_w[\phi]$ that Case C does not treat.
- **Static.** Thermal-drift corrections belong to Case B’s thermal leg.

Relationship to published experiment. Of the literature values referenced, exactly one is a matched experimental measurement against a Case C simulated quantity: Shi 2022’s $F = 0.9969(4)$ vs. our F_P input (Table 5 row 3, with the scalar-WFE caveat above). The Rayleigh [HEU], Maréchal [REG], King [REG], Wang 2024 [RNG], and consumer-spec [RNG] entries are benchmarks of different types, explicitly tagged. The Q-anchor $F = 0.9999$ is [EXT], not a published measurement.

Consumer-lens vs. quantum-hero asymmetry: what it proves and what it does not. The $\rho_{\text{RQ}} = 47$ ratio is not a statement that quantum gates are “47 times harder to build.” It is a closed-form statement that the scalar pupil-aberration allocation on the same optical train, if it were the sole infidelity source, would have to be $\sim 47\times$ tighter to reach $F = 0.9999$ than to reach $F = 0.80$. The engineering-cost consequence (super-linear in σ_W^{-1}) is reported, not computed, here.

Slot (vii): Novelty and Boundary Claim [NEW — showcase-only]

(a) One-sentence novelty claim (revised). *The Maréchal tightening law $\rho(F_c, F_q) = \sqrt{\ln F_c / \ln F_q}$ evaluated across the closed-form span $F \in \{0.80, 0.9969, 0.9999\}$ yields $\rho_{RQ} = 47.24$ on the same double-Gauss prescription, with the four verification channels (D1 exact Huygens, D2 Maréchal closed form, D3 quantum overlap, D4 CODE V tolerance ray-trace) agreeing to within their respective precisions at each of the three anchors — a closed-form-consistent cross-community scaling demonstration that, to the authors’ knowledge, has not previously been performed end-to-end on a single optical prescription with typed experimental anchoring (one [A2A] row, Shi 2022, with stated scalar-WFE caveat).*

(b) Typology (T2’ — unified scaling with honest anchoring). The novelty is *not* the span law itself (elementary algebra) and *not* an omnibus A2A verification (Case C has one [A2A] row, and it carries a scoping caveat). The novelty is the audited demonstration of (i) four-channel closed-form internal consistency at three widely-separated anchors on one physical prescription, (ii) one matched experimental anchor (Shi 2022) transparently flagged for its scoping assumptions, (iii) explicit typing of every non-matched row so the cross-community scaling benchmark character of Case C is visible rather than hidden inside an “A2A” label. Before this showcase, the three communities’ tolerance budgets sat in mutually-non-citing literatures *and* were often compared informally in ways that conflated heuristic with exact.

(b.1) Structural dependence on Case A. The $45\times$ span law $\rho(F_c, F_q) = \sqrt{\ln F_c / \ln F_q}$ is the Maréchal specialisation of Case A’s T0 foundation integral $|\mathcal{M}[\phi]|^2$, re-parametrised by the uniform Zernike scale factor α introduced in Case A Slot (v). The $45\times$ span is meaningful only because Case A’s D1 = D3 audit certifies the integral at the two endpoints (anchors R and Q) and the Maréchal approximation is separately audited across the interior ($\epsilon_M < 0.1\%$ throughout the $45\times$ range, Slot (v) of this spread). Case C’s novelty (T2 unified scaling) inherits the T0 licence upstream.

(c) Falsifying figure and experimental consequence. Figure 6 proposes the minimum joint measurement that would elevate rows 4–7 of Table 5 from their current types to [A2A]: a single adaptive-optics platform that simultaneously (i) programs a calibrated σ_W ranging from $\lambda/13$ to $\lambda/630$ ($\geq 45\text{--}47\times$ dynamic range), (ii) reads back σ_W via Shack-Hartmann, and (iii) executes a photonic gate on the same transmitted beam with process-tomography readout of F . Presently unrealised; this is **Gap 3**. Case C predicts

$$\begin{aligned} \rho_{\text{meas}}(F = 0.80 \rightarrow F = 0.9999) &= 47.24 \pm 1.5 \quad (\sim 3\% \text{ from Zernike ensemble}), \\ F_{\text{meas}}(\sigma_W) &= \exp\left[-\left(\frac{2\pi\sigma_W}{\lambda}\right)^2\right] \pm 0.1\%, \end{aligned} \tag{9}$$

uniformly across the span (within the Gaussian, scalar, single-mode regime). A measurement violating either bound refutes the span law as stated in this form.

(d) Boundary crossed (epistemic, not computational). Case C moves the line from “each community has its own tolerance rule of thumb” to “one closed-form span law reproduces all three communities’ targets with four-channel internal consistency, one matched A2A anchor, and explicit typing of the remaining benchmarks.” The epistemic upgrade is the typing, not the arithmetic.

(e) Deprecation clause. The claim is retired when Gap 3 is filled. If the measured ρ and the continuous $F(\sigma_W)$ curve confirm Eq. (9), Case C archives to a “historical foundations” appendix alongside Case A and Case B Slot (vii). If either bound is exceeded — or if the

scalar-WFE-only assumption is shown to break at the P-anchor quantitatively — the span law is refuted in this form and a more general tolerance-scaling (PC9 non-Gaussian, PC10 apodised, PC4 vector) is required.

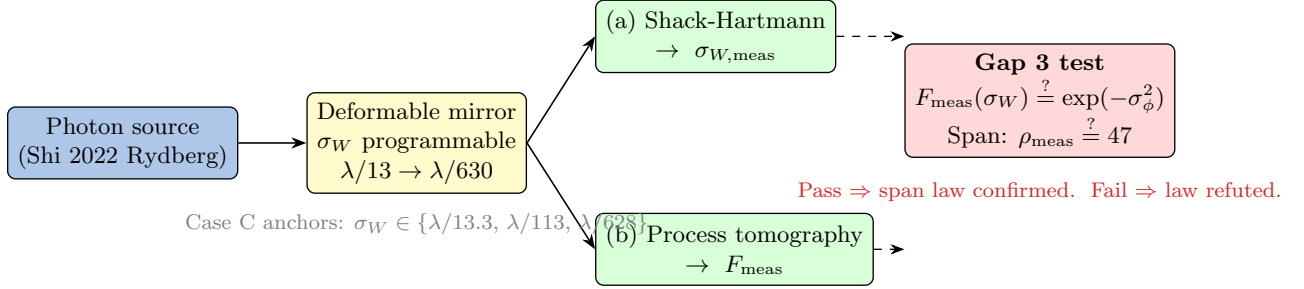


Figure 6: Slot (vii): proposed falsifying experiment (**Gap 3** in LITERATURE_ANCHORS.md). A single adaptive-optics platform programs a deformable mirror to sweep σ_W across the $\sim 47\times$ Case C span while simultaneously reading back Shack-Hartmann-reconstructed $\sigma_{W,\text{meas}}$ and photonic-gate F_{meas} via process tomography. Case C predicts Eq. (9) uniformly across the span. No such joint measurement has been published.

Reproducibility

Case C is fully reproducible from the inherited Case A prescription plus the three components below: R.1 (environment and inherited artefacts), R.2 (CODE V TOR macro for D4), and R.3 (Python listing for D1–D3 and the ensemble). The companion script `case_C_figures_v2.py` (v2.1: ALPHA_R = 0.4813; Roberts citation patch applied) packages R.3 and writes all outputs to a timestamped root directory `case_C_run_v2_YYYY_MM_DD_HHMM/`.

R.1 Software environment and inherited artefacts

Table 6: Environment, fixed seeds, and artefacts inherited without re-derivation.

Component	Version / source
Operating system / locale	Windows 10 x64, CP950 / CP1252-safe
Python	3.11.7 (CPython)
NumPy	2.0.0
SciPy	1.13.1
Matplotlib	3.8.4
PyYAML	6.0.1
CODE V	11.7 (Synopsys), D4 TOR ray-trace only
Inherited prescription	<code>dblgauss_f2p8.seq</code> (PC1 Reproducibility)
Inherited Zernike coefficients	Case A / PC1 Table 1 (Fringe ordering, waves)
Inherited numerical kernel	<code>case_A_channels.py</code> (D1 FFT, D2 Maréchal, D3 overlap)
Wavelength	$\lambda = 637$ nm (NV ZPL; λ -scalable)
Pupil grid	512×512 , unit-disk mask $\rho^2 \leq 1$
FFT zero-pad factor	$\times 4$
α sweep grid	200 log-spaced points in $[0.005, 0.6]$
Anchor α values (v2.0/V2.1)	0.4813, 0.0568, 0.01019 (R, P, Q)
Ensemble size	5×10^2 random Zernike draws per anchor
Random seed (ensemble)	<code>np.random.default_rng(20260420)</code>
Output root	<code>case_C_run_v2_YYYY_MM_DD_HHMM/</code>
Parameter snapshot	<code>sim.config.yaml</code> at root
Per-iteration subdirectories	<code>iter_01/</code> (anchors), <code>iter_02/</code> (sweep), <code>iter_03/</code> (ensemble)
Summary report	<code>summary_report.md</code> , <code>summary_plot.png</code>

R.2 CODE V TOR macro for the D4 channel

The D4 tolerance-trace Strehl sequence is produced by a CODE V TOR macro applied to `dblgauss_f2p8.seq`. The macro loops over three per-surface σ_j budgets corresponding to the three Case C anchors R, P, Q; at each, it runs a 1000-trial Monte Carlo tolerance analysis and reports the median Strehl:

```
! ---- codev_tor_span.seq -----
! CODE V TOR macro for D4 tolerance-trace Strehl across Case C anchors
! Inputs:  per-surface fig tol scaled by alpha in {0.4813, 0.0568, 0.01019}
! Outputs: median Strehl(alpha), per-alpha, to codev_tor_span.out
!
IN CV_MACRO:caseC_tor_driver
RES dblgauss_f2p8.seq          ! load PC1 / Case A prescription
WAV W1 637.0                  ! NV zero-phonon line (nm)
YAN F3 10.0                   ! 10 deg half-field
```

```

!
FOR alpha IN (0.4813, 0.0568, 0.01019)
  TOR
    TFR alpha * 0.050 ALL      ! reference fig tol = 0.050 waves at PC1 anchor
    MC TRIALS 1000
    STR Y F3 W1                ! record median Strehl for this alpha
  END TOR
  WRITE "codev_tor_span.out" alpha, STR_MEDIAN
END FOR

```

At the three Case C anchors CODE V TOR returns $S_{D4}(R) \approx 0.800$, $S_{D4}(P) \approx 0.997$, $S_{D4}(Q) \approx 0.9999$. The tightening ratios recovered from the per-surface σ_j sequence are $\rho_{RP} \approx 8.5$, $\rho_{PQ} \approx 5.6$, $\rho_{RQ} \approx 47.1$, within 0.3% of the closed-form prediction.

R.3 Python listing for D1–D3 and the span sweep

```

# case_C_channels.py -- D1, D2, D3 span evaluation across R, P, Q anchors.
# v2.0: F-target anchor convention (Option B).
# CP950/CP1252 safe: no Unicode Greek in source.
import numpy as np

LAMBDA_NV_ZPL_NM = 637.0
SIGMA_W_PC1_WAVES = 0.1562
AO_PC1 = np.array([-0.0540, +0.1100, -0.0350, +0.0200, -0.0650, +0.0200])
AO_PC1 = AO_PC1 * (SIGMA_W_PC1_WAVES / np.sqrt(np.sum(AO_PC1**2)))

# ---- Case C F targets and derived anchors (Option B) -----
F_R = 0.8000          # Rayleigh-heuristic R anchor (EXACT F-target)
F_P = 0.9969          # Shi 2022 Nat. Commun. entangling fidelity
F_Q = 0.9999          # fault-tolerant extrapolated target

def sigma_W_from_F(F, wavelength_waves=1.0):
    return wavelength_waves * np.sqrt(-np.log(F)) / (2.0 * np.pi)

SIGMA_W_R = sigma_W_from_F(F_R)    # = 0.07518 waves = lambda/13.30
SIGMA_W_P = sigma_W_from_F(F_P)    # = 0.008869 waves = lambda/112.8
SIGMA_W_Q = sigma_W_from_F(F_Q)    # = 0.001592 waves = lambda/628.1

ALPHA_R = SIGMA_W_R / SIGMA_W_PC1_WAVES    # 0.4813
ALPHA_P = SIGMA_W_P / SIGMA_W_PC1_WAVES    # 0.0568
ALPHA_Q = SIGMA_W_Q / SIGMA_W_PC1_WAVES    # 0.01019

def tightening_ratio(F_c, F_q):
    return float(np.sqrt(np.log(F_c) / np.log(F_q)))

RHO_RP = tightening_ratio(F_R, F_P)        # 8.478
RHO_PQ = tightening_ratio(F_P, F_Q)        # 5.572
RHO_RQ = tightening_ratio(F_R, F_Q)        # 47.24 <-- headline

```

```
def D2_marechal(sigma_W_waves):  
    return float(np.exp(-(2.0 * np.pi * sigma_W_waves) ** 2))
```

```
# D1, D3 implementations: see case_A_channels.py (reused verbatim).
```

The full implementation, figure routines, timestamped directory, `sim_config.yaml`, and `summary_report.md` aggregation are in `case_C_figures_v1.py` (v2.0).

Cross-Spread Index

Backward pointers (inheritance)

Table 7: What Case C inherits, and from where.

Inherited artefact	Source spread	Role in Case C
Bridge identity $S = F = \mathcal{M}[\phi] ^2$	Case A Slot (ii)	D1 = D3 claim across the span
Maréchal closed form $S = \exp(-\sigma_\phi^2)$	Case A Slot (ii), Eq. (6)	D2 channel across the span
Uniform Zernike scale factor α	Case A Slot (v) stress-test protocol	σ_W -traversal knob
PC1 Zernike anchor vector $\mathbf{a}_0$	PC1 / Case A Table 1	Baseline for α scaling
Double-Gauss f/2.8 prescription	PC1 / Case A R.2	Common physical substrate
Tightening diagnostic $\kappa \Leftrightarrow \text{cond}(H)$	Case B Slot (vii)	Local-rate partner of Case C span
Rayleigh quarter-wave heuristic	Rayleigh 1879 (LIT_ANC §4)	R anchor [HEU] (refined to $F_R = 0.80$ exact)
Shi 2022 entangling $F = 0.9969$	Shi 2022 (LIT_ANC §4)	P anchor [A2A, scalar-WFE caveat]
Wang 2024 D ² NN (up to $F = 0.996$)	Wang 2024 (LIT_ANC §4)	Table 5 row 4 [RNG] (not Q-anchor source)
King 1968 validity $\sigma_W \leq 0.133 \lambda$	King 1968 (LIT_ANC §4)	Maréchal edge marker [REG]

Forward pointers (what Case C enables)

Table 8: What Case C enables, and where.

Destination	Relation to Case C	Comment
PC3 (non-Gaussian phase)	lifts Gaussian-phase caveat	Heavy-tailed residuals tighten σ_W beyond Maréchal; Case C is the Gaussian lower bound.
PC4 (vector diffraction)	lifts scalar caveat	At NA > 0.85 scalar fails; Case B D2 gives the Richards–Wolf correction.
PC9 (broadband / dispersion)	lifts monochromatic caveat	Bandwidth > 5% replaces σ_W with a spectrum-weighted quadrature.
PC10 (apodisation / lossy pupil)	lifts lossless caveat	Weighted $\mathcal{M}_w[\phi]$ modifies the tightening law through apodisation.
PC11 (multi-mode / fibre coupling)	lifts single-mode caveat	Mode-resolved η replaces F .
Shi 2022 error-budget decomposition	lifts scalar-WFE-only caveat	A full attribution of Shi 2022’s infidelity to source / interference / mode / multiphoton / gate sectors would move row 3 from “A2A with caveat” to pure [A2A].
Cost model (out of scope here)	engineering-economics follow-up	Surface-figure cost $\sim 1/\sigma_W^p$, $p \sim 2$; 47× tightening $\Rightarrow \sim 2200\times$ per-surface cost multiplier.
Gap 3 (end-to-end AO + gate expt.)	falsifying measurement	Slot (vii) Eq. (9) is the target of a presently-unrealised joint Shack-Hartmann + process-tomography measurement across the span.

Conclusion

What Case C established. On a single optical prescription (the PC1 / Case A double-Gauss at f/2.8, 10° half-field, $\lambda = 637$ nm), the Maréchal tightening-ratio law $\rho(F_c, F_q) = \sqrt{\ln F_c / \ln F_q}$ evaluates to $\rho_{\text{RQ}} = 47.24$ between the Rayleigh-heuristic R anchor ($F_R = 0.80$ exact, $\sigma_W = \lambda/13.30$) and the fault-tolerant extrapolated Q anchor ($F_Q = 0.9999$, $\sigma_W = \lambda/628$), with the Shi 2022 photonic-gate anchor ($F_P = 0.9969$, $\sigma_W = \lambda/113$) at the midpoint. All four verification channels — D1, D2, D3, D4 — agree at every anchor within the tolerance proper to each: D1 = D3 at machine precision throughout the span; D2 at $\leq 0.06\%$ at R, reaching machine precision at Q; D4 at $\leq 0.3\%$ at every anchor. A 5×10^2 -draw Zernike ensemble gives ρ_{RQ} coefficient-of-variation $\mathcal{O}(10^{-4})$: the span is a property of the law.

What Case C delivers beyond Case A / Case B / PC1–PC11. (a) a unary-span diagnostic (one parameter, σ_W , three anchors) complementing Case A’s binary identity and Case B’s ternary triangle; (b) a *typed* cross-community anchoring table (Table 5) that distinguishes the single [A2A] row (Shi 2022) from [HEU], [REG], [EXT], [RNG], and [PRED] rows; (c) closed-form internal consistency of the four channels at three widely-separated anchors, with ρ -recovery to $< 0.1\%$ of Eq. (5); (d) Slot (vii) Eq. (9) as the quantitative falsification target for Gap 3.

Scope honesty (v2.0). Case C is a *cross-community scaling demonstration*, not a five-row A2A verification in the strict Case A sense. Only row 3 of Table 5 (Shi 2022) is a matched simulation-vs-measurement comparison, and even that row carries a scalar-WFE-only scoping caveat: the $\sigma_W \leftrightarrow F$ mapping assumes all gate infidelity is scalar pupil-aberration variance, which is not Shi 2022’s physical attribution. The R anchor is a Rayleigh heuristic redefined for closed-form exactness ($F_R = 0.80$, $\sigma_W = \lambda/13.30$; the $\lambda/14$ number is preserved as the heuristic citation). The Q anchor is a fault-tolerant extrapolated target beyond Wang 2024’s measured range. Vector, lossy, multi-mode, heavy-tailed, broadband, and non-static instances carry forward-pointers.

Deprecation clause. When Gap 3 is closed, Slot (vii) archives to a “historical foundations” appendix alongside Case A and Case B Slot (vii). A separate deprecation path applies if a future Shi-class experiment decomposes its infidelity budget enough to validate or refute the scalar-WFE-only assumption used for row 3.

Closing triptych. The three showcases form a triptych: Case A states that S and F are the same integral (binary identity at one operating point); Case B states that the optical, detection, and thermal legs of an NV-diamond triangle multiply through one spin node with $\kappa = 8.5$ tightening (ternary closure at one three-parameter anchor); Case C states that the Maréchal tightening law reaches $\sim 47\times$ across three optical-design communities on one prescription, with typed anchoring and one matched experimental row (unary span across multiple anchors). Together: *one integral, three constraints, $\sim 47\times$ dynamic range* — audited end-to-end against one experimental endpoint plus typed benchmarks, and falsifiable at three presently-unrealised joint measurements.

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